

Speaker-invariant suprasegmental temporal features in normal and disguised speech

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Received 18 December 2014; received in revised form 7 October 2015; accepted 8 October 2015

Available online 19 October 2015

Abstract

Cases of blackmailing and kidnapping often involve the perpetrator disguising his voice. In the present contribution, we examined the degree to which one exemplary form of voice disguise – the imitation of a foreign dialect – affects suprasegmental temporal features. Results of two production experiments and one perception test revealed high between-speaker and low within-speaker variability across the disguise condition. Results from the perception test further underlined the difficulty of the imitation task, as the majority of speakers were not accepted as native speakers by listeners of the target dialect. At the same time, the high rate of rejections of imitations as authentic speech indicates a high sensitivity and dialectal awareness on the part of the listeners. Findings suggest that those imitators who were accepted as native speakers may have been accepted partly because they succeeded in adjusting speaking rate and global intensity peak variability to that of the target dialect, amongst other factors.

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Keywords: Voice disguise; Suprasegmental temporal features; Prosody; Forensic phonetics; Speaker invariance

1. Introduction

Voice disguise refers to the voluntary act of changing one's voice as a means to mask identity. In some forensic cases, perpetrators disguise their voices. The German Federal Police Office (BKA) reports that cases involving voice disguise range between 12.6% (Wagner, 1998) and 22.9% (Braun, 2006). Current research distinguishes between two types of voice disguise: electronic voice disguise, e.g. a perpetrator applies an electronic voice changer, and non-electronic voice disguise (Masthoff, 1996; Künzel, 2000; Perrot et al., 2007). Existing research on the former type recognizes the critical role of electronic voice disguise on voice identification: an artificial change in voice

fundamental frequency, for example, negatively affects identification performance (Clark and Foulkes, 2007). Non-electronic disguise, however, occurs more frequently and can typically be broken up into (a) a change in voice source characteristics (e.g. raised pitch, lowered pitch, whispered speech), (b) a change in resonance features (e.g. foreign object in vocal tract, hypo-/hypernasality, tissue held to mouth), (c) a change in language, accent, and dialect (Künzel, 2000; Perrot et al., 2007; Neuhauser, 2011a).

Previous research has shown that voice disguise can lead to high within-speaker variability, affecting various acoustic parameters. Hollien and Majewski (1977) reported high within-speaker variability in long-term spectra under various voice disguise conditions (disguise of the speaker's own choice). Face covering as a means of voice disguise has been reported to affect spectral characteristics of fricatives (Fecher and Watt, 2011) and the amplitude of

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the overall signal (Llamas et al., 2009). Hirson and Duckworth (1993) showed that the intentional application of creaky voice as a disguise affects spectral characteristics of /s/. A further strategy for disguising one's voice is to imitate another speaker. Eriksson and Wretling (1997) and Endres et al. (1971) reported that speakers particularly aim at matching f0 and formant frequencies when asked to imitate another speaker. Most often, this f0 and formant matching is successful. High within-speaker variability makes it difficult for forensic practitioners to draw conclusions about two samples coming from the same speaker. It is thus desirable in forensic cases that the acoustic features examined reveal maximum between-speaker and minimum within-speaker variability (Nolan, 2009).

How suprasegmental temporal features behave in voice-disguised speech is currently largely unknown. The objective of this contribution is to test if and how temporal features that are not restricted to a single segment but rather to the more global temporal organization of speech remain speaker-invariant in one form of disguised speech, namely in cases where a different dialect is imitated. Such features have shown promising trends in this respect: preliminary research in the context of voice disguise has shown that suprasegmental temporal features such as the amount of voicing in the speech signal are affected only very little by articulatory obstructions (Hove and Dellwo, 2012). Moreover, the percentage over which speech is vocalic remained relatively speaker-specific even in dialect imitation tasks (Dellwo et al., 2009). In the context of forensic speaker comparison in undisguised speech, Leemann et al. (2014a) reported suprasegmental temporal features to demonstrate high between-speaker variability in both read and spontaneous speech and little within-speaker variability across speaking styles and different signal qualities. Farrús et al. (2008) reported results from voice imitation tasks which underlined that durational features – such as word-internal voiced or unvoiced portions – are particularly difficult to imitate. There are a handful of other studies that were not conducted in the context of forensic phonetics but which, too, revealed a high degree of speaker-individuality in suprasegmental temporal features (Laan, 1997; Dellwo, 2010; Wiget et al., 2010; Yoon, 2010; Arvaniti, 2012; Dellwo et al., 2015).

In the present study we tested one type of voice disguise, namely the imitation of another dialect. This is a realistic task as Markham (1999) has shown that speakers are well able to imitate different dialects – albeit with different degrees of accuracy. We recorded speakers in two sessions: in the first session, the speakers read sentences in their native Zurich German dialect. In the second session, they read the same sentences in their best imitation of a Bern dialect. Swiss German listeners are well aware of dialectal variation, and perform well, for example, in dialect identification tasks: Leemann and Siebenhaar (2008) and Guntern (2011) reported that naïve listeners can accurately identify a speaker's dialect in four-alternative and eight-alternative forced-choice experiments with a recognition

rate of 86% and 74% respectively (both experiments included Bern and Zurich German). Zurich and Bern German are perceived differently in society. Bern German is Switzerland's most popular dialect (Werlen, 1985) and characterized as 'cozy', 'slow', and 'colorful'; Zurich German, on the other hand, is viewed as 'modern', 'fast', and 'vulgar' (Ris, 1992; Hengartner, 1995; Werlen, 1985; Berthele, 2006). Because of its slowness in particular (cf. Leemann, 2012; Leemann et al., 2014d), Bern German is frequently caricatured in stand up comedy and TV sketches.

Segmentally and prosodically, the two dialects differ from one another. Bern German has, for example, mainly lowered its vowels from Middle High German; Zurich German, on the other hand, has raised them or they remained the same (Haas, 1985; Marti, 1985; Fleischer and Schmid, 2006). Bern German further features lax near-close near-front vowels /ɪ/ and /y/ and a near-back lax vowel /ʊ/, where Zurich German has close vowels. A major phonological difference is Bern German /l/-vocalization in some phonological contexts (Leemann et al., 2014c). Prosodically, Bern German is characterized by slower speaking rates than Zurich German (Leemann, 2012; Leemann et al., 2014d) and delayed pitch peaks (Leemann, 2012). Bern German is further reported to have a lower percentage of vocalic intervals (%V, cf. Ramus et al., 1999) and a lower rate-normalized consonantal pairwise variability index (nPVI_C, cf. Grabe and Low, 2002) as shown on a small corpus of read speech (Leemann et al., 2012). Possible reasons for between-dialect differences in suprasegmental temporal measures are reported in Leemann et al. (2014b): On the one hand, Bern dialect, for example, realizes nominative definite articles as [dər] contrasting with Zurich German [də]. This phenomenon may increase the number of CVC syllables in Bern German, which may lead to less variability in the consonantal pairwise variability index. Secondly, two relatively prominent phonological processes are likely to contribute to differences in suprasegmental temporal patterns between the two dialects: (i) Bern dialect-typical /nd/-velarization causes a change in syllable structure between the two dialects ('Hund', *dog*, realized as [hʊŋ] CVC in Bern German and as [hʊnd] CVCC in Zurich German); (ii) /l/-vocalization in Bern German (cf. Leemann et al., 2014c) leads to differences in syllable structures, such as in 'Milch', *milk*, realized as [mɪʊx], CVC in Bern German and as [mɪlx] CVCC in Zurich German). Such differences in syllable structure are likely to translate to differences in the rhythmic structure (cf. Dauer, 1983): dialects that exhibit less complex clusters in some phonological contexts, e.g. Bern German in the examples above, are likely to exhibit smaller variation in the respective suprasegmental temporal features.

The main aim of this study is to examine the effect of dialect imitation – Zurich German speakers imitating Bern German – on suprasegmental temporal features. Given the evidence reported earlier, we hypothesize to find some

suprasegmental temporal features to remain speaker-invariant across voice disguise. To this end, we conducted three experiments:

Experiment I: In the first production experiment we tested the variability of automatically retrievable suprasegmental temporal features between Bern and Zurich German. To this end we analyzed a corpus of native Bern and Zurich German speech (corpus use courtesy of Volker Dellwo). With this production experiment we can investigate whether the imitation of Bern German by Zurich German speakers does in fact introduce within-speaker variability.

Experiment II: In the second production experiment we examined the variability of suprasegmental temporal measures between the disguised and the non-disguised speech. We recorded native Zurich German speakers in two sessions: in the first session, they read sentences in their native dialect. In the second session, they read the same sentences in their best imitation of the Bern dialect, i.e. a disguise condition. We automatically measured the same features as used in Experiment I.

Experiment III: In a perception experiment, native Bern German listeners rated the disguised speech from Experiment II for authenticity. Subsequently, the suprasegmental temporal features of those speakers who produced the most authentic Bern German in the imitation task were compared to those who were not perceived as authentic.

Sections 2–4 cover Experiments I, II, and III respectively. Results of all three experiments are discussed in Section 5.

2. Experiment I

2.1. Speakers

16 speakers (eight Zurich German/eight Bern German) were recorded in Zurich and in Bern. Eligibility criteria required individuals to demonstrate little regional dialect variability. Three females and five males per dialect were recorded. Speakers were between 20 and 30 years old. None of the speakers reported hearing or speech disorders. Recordings were conducted with a Zoom H2 recorder (sampling rate of 44.1 kHz; 16 bit quantization).

2.2. Material

Subjects were asked to read the Bamford–Kowal–Bench (BKB) sentences (Bench et al., 1979). These 191 sentences were translated from their original English to Zurich German and Bern German. As there is no standard writing system for Swiss German dialects, the transcripts were tailored specifically for Zurich German and Bern German. Transcripts were constructed so they reflected a balance between writing the sentences as phonetically as possible

while at the same time following a spelling close to Standard German (see Appendix A). These 3056 sentences (191 sentences * 16 speakers) constituted the corpus for Experiment I.

2.3. Data editing

To prepare the data for the application of automatic suprasegmental temporal measures (cf. 2.4), trained phoneticians labeled each signal for silent parts – i.e. breathing pauses and silent hesitations longer than 200 ms – in Praat (Boersma and Weenink, 2014).

2.4. Temporal measures applied

In the present study, we applied automatic suprasegmental temporal measures based on amplitude peaks and voicing features of the speech signal. These features, annotated as amplitude-peak-to-amplitude-peak intervals and voiced or voiceless intervals, allow for the derivation of (a) speaking-rate-normalized global interval variability measures – captured by the variation coefficient measures (see below), of (b) speaking-rate-normalized local interval variability measures – captured by the pairwise variability indices, and of (c) overall voicing in the signal – captured by the percentage of voiced portions in the speech signal. Previous studies have shown that these measures reveal a high degree of between-speaker variability and at the same time within-speaker robustness towards speaking style, channel and tempo variability and are thus promising for forensic phonetics (Leemann et al., 2014a; Dellwo et al., 2012, 2015). From a more linguistic typology strain of research, Dellwo (2006) has shown the usefulness of speaking rate-normalized global interval variability measures in distinguishing languages into different rhythm classes. Similarly, Grabe and Low (2002) provided evidence that local interval variability measures can be used to categorize languages rhythmically. Finally, Dellwo et al. (2007) and Dellwo and Fourcin (2013) reported a classification of languages based on voicing measures (i.e. the percentage over which speech is voiced and the variability of voiced intervals in the signal) alone. Such measures do not require segment-by-segment labeling.

2.4.1. Peak-to-peak measures

We applied two measures that are grounded in P-center theory (Morton et al., 1976; Fowler, 1979; Marcus, 1981). Amongst other acoustic correlates, P-center theory holds that perceptually prominent centers of the speech signal exist in the vicinity of vowel intensity peaks. There are alternative models, however, that attribute a more critical role to vowel onsets in terms of the perception of rhythmic beats (Allen, 1972). Both models, however, thus suggest one such perceptually prominent center per syllable. We used a Praat (Boersma and Weenink, 2014) script that detects amplitude maxima in the amplitude envelope (<10 Hz; cf. <http://www.pholab.uzh.ch/en/leute/dellwo/software.html>;

Dellwo et al., 2012). We then identified the intervals between the amplitude peak points (Fig. 1, tier three). The interval between the onset of speech and the first peak point as well as the interval between the last peak point and the offset of speech were excluded from the subsequent analysis.

The Zurich German phrase depicted in Fig. 1 reads: “Ich han käi äigeni Band” – *I do not have a band myself* (cf. Appendix B, sentence (2)). Tier one shows the segments, tier two the syllables, tier three the automatically detected amplitude peaks (peak-to-peak intervals are numbered consecutively), and tier four the automatically annotated voicing portions (cf. (2) *Speech voicing measures* below). In Fig. 1, tiers one and two are annotated for demonstration purposes only, i.e. to indicate the relationship between automatically detected peaks (tier three) and syllables (tier two). In many instances, the number of syllables corresponds to the number of automatically detected peaks. In the phrase depicted in Fig. 1, this is the case for intervals 1–6. Intervals 7 and 8 are, in the syllabic sense (or in the P-center theory sense), “erroneously” annotated. This margin of “error” is one of the – tolerable – trade-offs when applying automatic measures. Informal tests on TEVOID data (Dellwo et al., 2012, 2015; Leemann et al., 2014a) indicated a high correlation ($r > .9$) of the number of automatically detected amplitude peaks and the number of syllables. Our primary goal is to find measures that reveal speaker-idiosyncrasy. Peak-to-peak measures have been shown to fulfill this criterion (cf. Leemann et al., 2014a). Of course it would further be ideal if the measures used fully translate to linguistic units, such as the syllable. But, given that the measures used in fact reveal speaker idiosyncrasy, the error rate reported here would not affect forensic casework. It simply means that, conceptually, there is a small margin of error in automatic peak detection if detecting one peak per syllable is wished for – speaker-specific patterns remain the same regardless.

Based on these interval durations between amplitude peaks we derive the following two suprasegmental characteristics of the speech signal (Dellwo et al., 2012):

- (A) The rate-normalized standard deviation of peak-to-peak interval durations *varcoPeak* (Dellwo et al., 2012):

$$\text{varcoPeak} = 100 \cdot \frac{\Delta_{\text{peak-to-peak}}}{\overline{\text{peak-to-peak}}}$$

$\Delta_{\text{peak-to-peak}}$ is the standard deviation of peak-to-peak interval durations and $\overline{\text{peak-to-peak}}$ the mean peak-to-peak interval duration.

- (B) The rate-normalized average differences between consecutive peak-to-peak interval durations *nPVI_{peak}* (Dellwo et al., 2012):

$$\text{nPVI}_{\text{Peak}} = 100 \cdot \left[\frac{\sum_{k=1}^{m-1} \left| \frac{d_k - d_{k+1}}{(d_k + d_{k+1})/2} \right|}{m - 1} \right]$$

m is the number of peak-to-peak intervals and d_k the duration of the k th interval.

2.4.2. Speech voicing measures

We applied three measures that are based on speech voicing of the signal (Dellwo et al., 2007). Voicing detection was performed automatically using pitch detection in Praat (Boersma and Weenink, 2014) (standard settings). Voiced intervals in Fig. 1 are coded as “v”, unvoiced intervals as “u” in tier four. Altogether, five voiced intervals were detected in the phrase in Fig. 1. Based on this labeling of voicing we derived the following three measures (Dellwo et al., 2007; Dellwo and Fourcin, 2013):

- (C) The percentage over which speech is voiced %VO.
(D) The rate-normalized standard deviation of voiced interval durations *varcoVO*:

$$\text{varcoVO} = 100 \cdot \frac{\Delta VO}{\overline{VO}}$$

ΔVO is the standard deviation of voiced interval durations and $\overline{VO}$ the mean voiced interval duration.

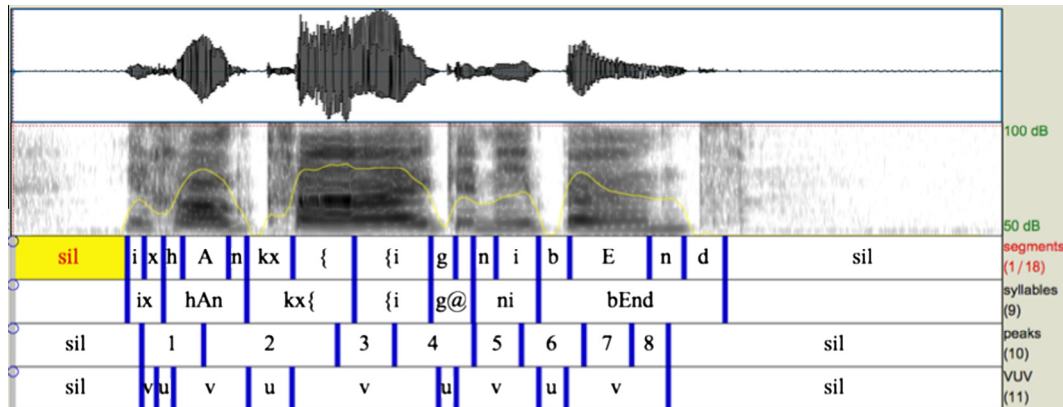


Fig. 1. Automatic peak detection (tier three) and automatic voicing detection (tier four).

- (E) The rate-normalized average differences between consecutive voiced interval durations $nPVI_VO$:

$$nPVI_VO = 100 \cdot \left[\frac{\sum_{k=1}^{m-1} \left| \frac{d_k - d_{k+1}}{(d_k + d_{k+1})/2} \right|}{m - 1} \right]$$

m is the number of voiced intervals and d_k the duration of the k th interval.

Temporal measures were calculated sentence-by-sentence, using durationAnalyzer.praat (available at: <http://www.pholab.uzh.ch/en/leute/dellwo/software.html>).

2.5. Statistical analyses

All data were analyzed using R (R Core Team, 2013) and the R package *lme4* (Bates and Maechler, 2009) as well as JMP (1989–2010). If not indicated otherwise, we analyzed data using linear mixed effect models (LMEs). Normality was checked by visual inspection of quantile plots. *Dialect* and *gender* were treated as fixed effects, *sentence* and *speaker* as random effects. We included by-speaker random slopes on the effect of *dialect*. Effects were tested by model comparison between a full model (e.g. `model_full <- lmer(percentVoiced ~ gender + dialect + (dialect|speaker) + (1|sentence), data = data)`) and a reduced model that excluded the effect to be tested. p -values were obtained by comparing the results from the two models using standard likelihood ratio tests (R code: `anova(model_full, model_reduced)`). For the assessment of the relative goodness of fit, we indicate AIC (Akaike Information Criterion) values. AIC is defined as:

$$AIC = 2k - 2\ln(L)$$

k stands for the number of parameters in the model and L represents the maximized value of the likelihood function for that model. AICs decrease with the goodness of fit of the model (Kliegl et al., 2011). We decided on an α level of .05.

2.6. Results

Table 1 summarizes the results obtained for each suprasegmental temporal measure.

Fig. 2 shows boxplots of the Bern German speakers' (BE, green) and Zurich German speakers' (ZH, purple) suprasegmental temporal measures. Non-overlapping notches indicate a significant difference between the medians in the pair.

The model with *dialect* as a fixed effect showed an improved goodness of fit in one out of five suprasegmental temporal measures. For varcoPeak, between-dialect differences were significant. Only in varcoPeak do the Zurich German ($M = .472\%$, $SD = .121$) speakers demonstrate significantly higher values than Bern German speakers ($M = .468\%$, $SD = .124$). However, Zurich German speakers descriptively also demonstrated higher nPVI_peak

Table 1

Summary of the statistics of the mixed model comparisons for *speaker* and *dialect* for each measure.

Temporal measure	Factor	Result
varcoPeak	Speaker	$p < .0001^*$, AIC = -4686
	Dialect	$p = .027^*$, AIC = -4686
nPVI_peak	Speaker	$p = .023^*$, AIC = 24,214
	Dialect	n.s.
%VO	Speaker	$p < .0001^*$, AIC = 21,229
	Dialect	n.s.
varcoVO	Speaker	$p < .0001^*$, AIC = 1174
	Dialect	n.s.
nPVI_VO	Speaker	$p < .0001^*$, AIC = 28,262
	Dialect	n.s.

values ($M = 50$, $SD = 14.1$) than Bern German speakers ($M = 49.1$, $SD = 14.1$), higher %VO values (ZH: $M = 60.7\%$, $SD = 10.6$; BE: $M = 59.1\%$, $SD = 12$), higher varcoVO values (ZH: $M = .65\%$, $SD = .23$; BE: $M = .62\%$, $SD = .22$), as well as higher nPVI_VO values (ZH: $M = 75.9$, $SD = 28.1$; BE: $M = 73.6$, $SD = 28$).

Table 1 further reveals significant between-speaker differences in all measures examined, which is in line with previous findings on between-speaker differences in these measures (cf. Leemann et al., 2014a). Speaker-specific patterns are explored in Fig. 3.

Fig. 3 shows that the given set of Bern and Zurich German speakers differ from one another only slightly while, at the same time, between-speaker variability is high in each dialect. This accounts for the non-significant dialect effects but significant speaker effects reported for four out of five measures in Table 1. One speaker in particular, BE13, affects the overall mean for the Bern German group. This is especially true for %VO ($M = 72.5\%$, $SD = 13.2$), varcoVO ($M = .75\%$, $SD = .31$), and nPVI_VO ($M = 93.9$, $SD = 40.1$). If BE13 is excluded from the Bern German set, between-dialect differences become nearly significant for nPVI_VO ($p = .11$) and %VO ($p = .24$). For varcoVO, dialects would differ from one another if speaker BE13 were removed ($p < .05^*$).

The aim of Experiment I was to examine if Bern and Zurich German differ in the automatic measures applied. With this production experiment, we were able to test whether the imitation of Bern German by Zurich German speakers (see Experiment II) in fact introduced within-speaker variability. Results showed significant between-dialect differences only in one out of five suprasegmental temporal measures. Including *speaker* as a random effect in the statistical model – as we have done now – is, however, a conservative way of testing between-dialect differences, as we want to generalize over an entire speaker population. If we were to treat *speaker* more liberally and include this factor as a fixed effect, varcoPeak differences become non-significant, whereas dialects would differ in nPVI_peak, varcoVO, and nPVI_VO (all $p < .0001^*$). This of course would mean that the tested between-dialect differences are only valid for this exact set of Bern and Zurich German speakers.

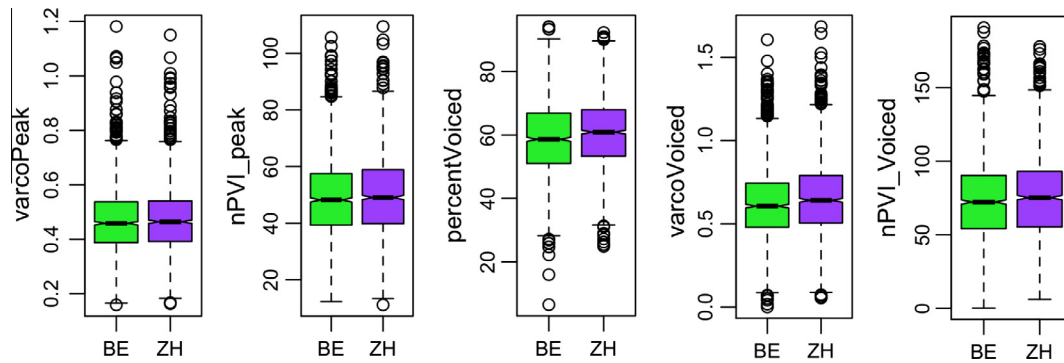


Fig. 2. Boxplots of the two dialects' suprasegmental temporal measures for Bern German (BE, green) and Zurich German (ZH, purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Experiment II

3.1. Speakers

For Experiment II, a new group of ten speakers of Zurich German (five male/five female) were recorded at the Phonetics Laboratory, University of Zurich. Speakers were University of Zurich students and showed little regional dialect variability in their native Zurich German, as attested by informal listening tests and by a questionnaire on their growing up and their parents' dialects. Most of them grew up in the Zurich city area and all of them had at least one parent who speaks Zurich German as a native language. All speakers indicated that they learned two or more second languages. They are thus aware of linguistic variation in the broader sense. As mentioned earlier, most Swiss German speakers are aware that Zurich and Bern German are perceived differently in society. In particular, they know that the most salient feature of Bern German is its slowness, and because of that, it is frequently caricatured on television or on the radio. It is further very likely that the speakers are aware of segmental differences, such as /l/-vocalization or more open vowels in Bern German (cf. Section 1). None of the speakers reported speech or hearing problems. The speakers were recorded in a sound-treated booth using an omnidirectional Earthworks QTC40 high definition condenser microphone (sampling rate of 44.1 kHz; 16 bit quantization).

3.2. Material

Each speaker read 74 Zurich German sentences from the TEVOID corpus (Dellwo et al. 2012, 2015; Leemann et al., 2014a). Sentences typically included 15–20 syllables and were written in Zurich German. The same sentences were then transliterated to Bern German and read by the subjects for the Bern dialect imitation condition (see Appendix B). Subjects were given the transcripts beforehand to prepare reading the sentences before the recording session. These 1480 sentences (74 Zurich German sentences * ten speakers + 74 Bern German sentences * the same ten speakers) constituted the voice disguise corpus of this study.

3.3. Data editing

To prepare the data for the application of automatic suprasegmental temporal measures (cf. Section 2.4), trained phoneticians (both authors) labeled each signal for silent parts in Praat (Boersma and Weenink, 2014).

3.4. Temporal measures applied

The same measures as described in Section 2.4 were applied.

3.5. Statistical analyses

LMEs were computed as described in Section 2.5. Our full model included *gender* and *disguise* as fixed effects; *speaker* as a random effect including by-*speaker* random slopes on the effect of *disguise*; and *sentence* as a random effect (e.g. R code: `modelVD_full <- lmer(varcoPeak ~ gender + disguise + (disguise|speaker) + (1|sentence), data = data)`). To test for *speaker* effects, the reduced model was coded as `(modelVD_reduced_speaker <- lmer(varcoPeak ~ gender + disguise + (1|sentence), data = data))`. To test for the effect of *disguise*, the reduced model was coded as `(modelVD_reduced_disguise <- lmer(varcoPeak ~ gender + (1|speaker) + (1|sentence), data = data))`. To test for the interaction *speaker* * *disguise*, we compared the full model with a reduced model which included the terms of the full model except for the random slope: *speaker* was included as random intercept (R code: `modelVD_reduced <- lmer(varcoPeak ~ gender + disguise + (1|speaker) + (1|sentence), data = data)`).

3.6. Results

Table 2 summarizes the results obtained.

Fig. 4 shows the boxplots of disguised (vdbe – 'voice disguise Bern', yellow) and non-disguised sentences' (ZH, purple) suprasegmental temporal measures.

When comparing these results with the ones in Experiment I, we find an opposite trend: the (authentic) Zurich variety in all instances shows lower values for our suprasegmental temporal measures, i.e. a lower overall

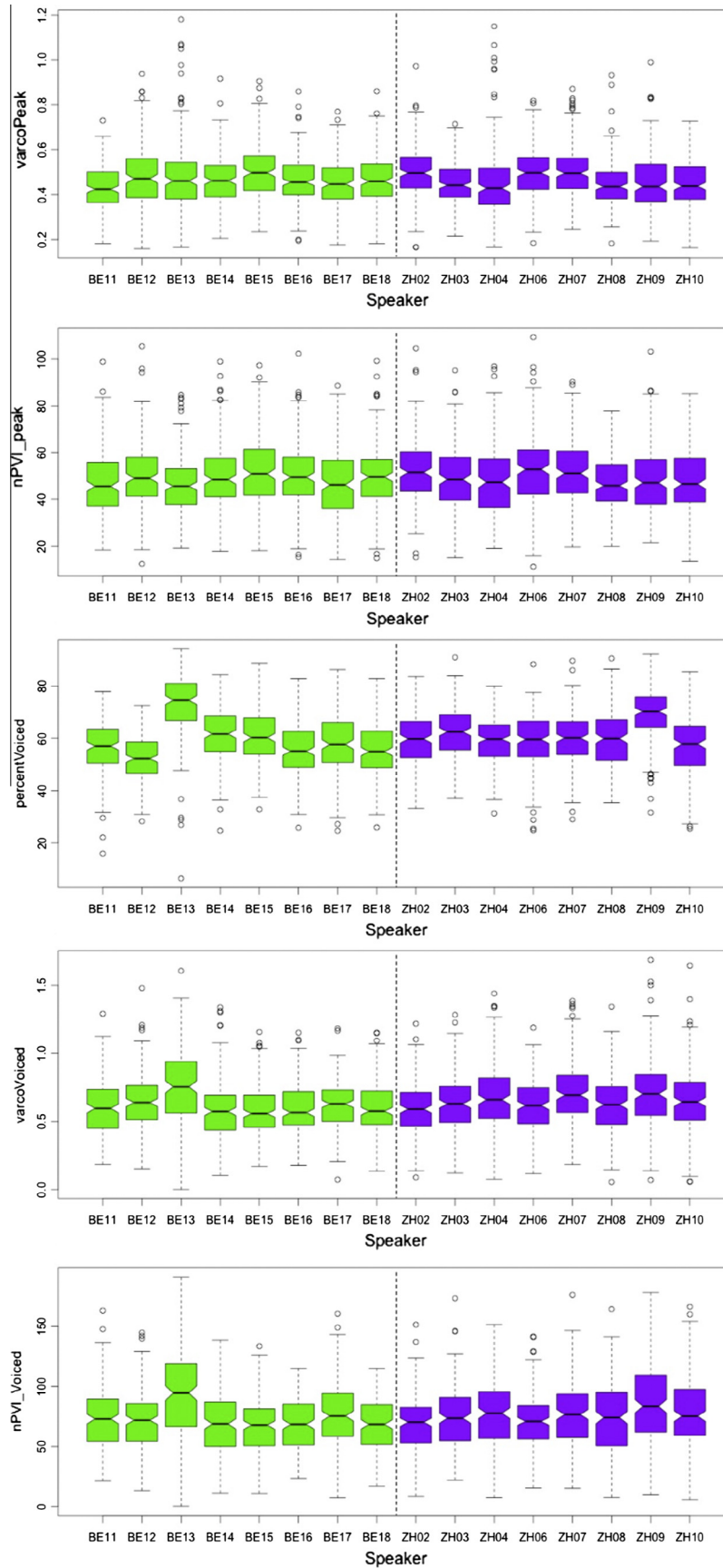


Fig. 3. Boxplots of Bern (green) and Zurich German (purple) suprasegmental temporal measures by speaker. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Summary of the statistics of the mixed model comparisons for *speaker*, *disguise*, and *speaker * disguise* for each measure.

Temporal measure	Factor	Result
varcoPeak	Speaker	$p < .0001^*$, AIC = -2680
	Disguise	$p < .0001^*$, AIC = -2680
	Speaker * disguise	n.s.
nPVI_peak	Speaker	$p < .0001^*$, AIC = 11,150
	Disguise	$p < .0001^*$, AIC = 11,150
	Speaker * disguise	n.s.
%VO	Speaker	$p < .0001^*$, AIC = 9180
	Disguise	$p < .0001^*$, AIC = 9180
	Speaker * disguise	$p < .0001^*$, AIC = 9180
varcoVO	Speaker	n.s.
	Disguise	n.s.
	Speaker * disguise	n.s.
nPVI_VO	Speaker	$p = .0001^*$, AIC = 12,742
	Disguise	n.s.
	Speaker * disguise	n.s.

percentage over which speech is voiced and smaller variability in voiced interval durations as well as in peak-to-peak interval durations.

varcoPeak: The model with *speaker* as random effect provided an improved goodness of fit, i.e. between-speaker differences were significant (cf. Table 2). Results further revealed that the disguised speech overall exhibits a significantly higher varcoPeak ($M = .42$, $SD = .1$) than non-disguised speech ($M = .39$, $SD = .11$). There was no significant interaction of *speaker * disguise*.

nPVI_peak: The model with *speaker* as random effect provided an improved goodness of fit, i.e. between-speaker differences were significant (cf. Table 2). Results further showed that the disguised speech overall exhibits a significantly higher nPVI_peak ($M = 44.9$, $SD = 11.9$) than non-disguised speech ($M = 40.8$, $SD = 11.2$). There was no significant interaction of *speaker * disguise*.

%VO: The model with *speaker* as random effect provided an improved goodness of fit, which entails that between-speaker differences were significant (cf. Table 2). Results indicated a significant difference in %VO across

the two conditions, with the disguised speech showing an increased %VO value (vdbe: $M = 65.1\%$, $SD = 7.8$, ZH: $M = 64\%$, $SD = 7.9$). There was a significant interaction of *speaker * disguise*. To test for *speaker* as a simple effect, we computed one LME on the disguised sentences and one LME on the non-disguised sentences. Both LMEs showed significant effects of *speaker* (Bonferroni-adjusted for *disguise*, $\alpha = 0.025$; disguised: $p < .0001^*$, AIC = 4562; non-disguised: $p < .0001^*$, AIC = 4623). To test for the simple effect of *disguise*, we again computed LMEs. For five out of ten speakers, there were no significant differences between disguised and non-disguised speech (Bonferroni-adjusted for *speaker*, $\alpha = 0.005$). For five speakers, there was a significant difference (speaker 5: $p = .0001^*$, AIC = 987; speaker 6: $p = .0003^*$, AIC = 949; speaker 7: $p < .0001^*$, AIC = 994; speaker 15: $p < .0001^*$, AIC = 1005; speaker 16: $p < .0001^*$, AIC = 955).

varcoVO: For varcoVO, neither *speaker* nor *disguise* generated significant effects.

nPVI_VO: For nPVI_VO, we observed a significant effect for *speaker*, whereas the effect of *dialect* and the interaction of both effects were not significant. Fig. 5

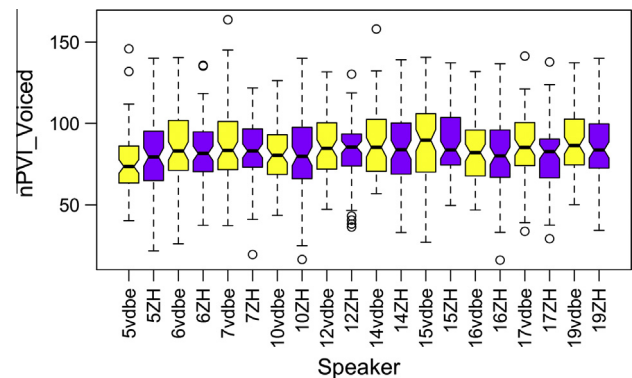


Fig. 5. Boxplots of the ten speakers' nPVI_VO values: disguised speech (vdbe, yellow) and non-disguised speech (ZH, purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

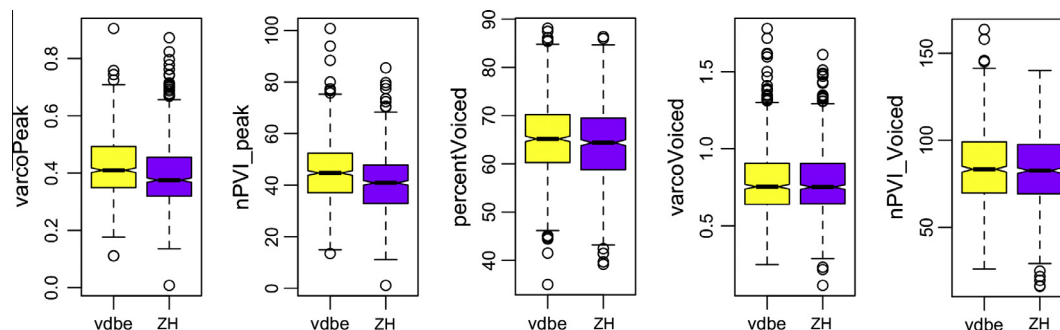


Fig. 4. Boxplots of the suprasegmental temporal measures for disguised speech (vdbe, yellow) and non-disguised speech (ZH, purple). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

shows the boxplots of the speakers' nPVI_VO values in disguised speech (yellow) and non-disguised speech (purple).

The boxplots reveal between-speaker differences and at the same time within-speaker invariance: speaker 5, for example, shows relatively low nPVI_VO values in both conditions (disguised: $M = 75.9$, $SD = 18.5$; non-disguised: $M = 79.5$, $SD = 21.8$) while speaker 15 shows relatively high values in both conditions (disguised: $M = 89.2$, $SD = 26$; non-disguised: $M = 89.8$, $SD = 20.9$). The model with *speaker* as random effect provided an improved goodness of fit, i.e. between-speaker variation was significant (cf. Table 2). All variation is caused by the factor *speaker*.

4. Experiment III

4.1. Speakers

Experiment III included three groups of speakers:

- Group (a) The same speakers as in Experiment II (ten Zurich German speakers imitating Bern German (cf. Section 3.1)).
- Group (b) Five newly recorded speakers of Bern German (one male/four females).
- Group (c) Five speakers of Zurich German (two males/three females) taken from the 16-speaker TEVOID corpus (Dellwo et al., 2012, 2015; Leemann et al., 2014a).

Speakers of groups (b) and (c) were also recorded at the Phonetics Laboratory, University of Zurich. German-speaking Switzerland is known to have distinct dialectal variation in a small and confined area, which is why we selected speakers of Bern German (b) and Zurich German (c) that demonstrated little regional variability in their respective dialects, i.e. they were speakers of Bern city and Zurich city. Both authors are dialectologists and screened speakers for dialectal coloring. None of the speakers reported speech or hearing problems. Recordings were conducted in a sound-treated room using an omnidirectional Earthworks QTC40 high definition condenser microphone (sampling rate of 44.1 kHz; 16 bit quantization).

4.2. Material

For speaker group (a), 15 of the 74 voice disguised sentences from Experiment II were randomly selected (cf. 3.2; see Appendix B). For speaker groups (b) and (c), the same material as in Experiment II was recorded and the same 15 sentences as for speaker group (a) were selected. We thus used 300 sentences in total as stimuli for the perception experiment (15 voice disguised sentences * ten speakers + 15 Bern German sentences * five speakers + 15 Zurich German sentences * five speakers).

4.3. Listeners

We tested a total of 22 (nine females/13 males) Swiss German listeners. All listeners were native speakers of Bern Swiss German with little regional dialect variability. The listeners were students of general linguistics at the University of Bern. As any Swiss German speaker they are familiar with the broader dialectal regions in Switzerland, including Zurich German. We did not further test their knowledge of different dialects. Furthermore, as any Swiss German speaker, they have learned English and French at school. None of the subjects reported significant problems with hearing or sight.

4.4. Procedure

Listeners were tested in a quiet room at the University of Bern. The experiment lasted between 15 and 25 min. The order of the stimuli was randomized separately for each subject. Each stimulus, i.e. each sentence, was presented over high quality headphones. Following stimulus presentation, subjects had to decide in a two-alternative forced choice setting whether the preceding sentence was spoken by a Bern German speaker. They gave their response by clicking on the corresponding button on a laptop computer. Listeners further indicated the confidence of their response on a three-point scale (sure, insecure, only guessing). The experiment interface is shown in Fig. 6.

4.5. Statistical analyses

Listener's response to each stimulus was coded as either '1' (the stimulus was accepted as coming from a Bern speaker) or '0' (the stimulus was rejected as coming from a Bern speaker). Because we used a two-alternative forced choice framework, chance level is 50%. We calculated a logistic mixed effects regression model. To test for the effect of *condition* (with the levels of Bern dialect, Zurich dialect, and imitated Bern dialect), *condition*, *speaker gender* and *listener gender* were treated as fixed effects whereas *speaker*, *sentence*, and *listener* were coded as random effects in the full model (R code: `model_full <- glmer(bernResponded ~ condition + genderListener + genderSpeaker + (1|speaker) + (1|sentence) + (1|listener), family = binomial(), data = data)`). The reduced model for *condition* was coded as `model_reduced <- glmer(bernResponded ~ genderListener + genderSpeaker + (1|speaker) + (1|sentence) + (1|listener), family = binomial(), data = data)`. Testing for the effects of *speaker* and *listener*, model comparisons were conducted analogously. Confidence rating spans from 1 (very confident) to 3 (only guessed).

4.6. Results

Fig. 7 shows the bar chart of the listeners' responses by *condition*, where the y-axis shows, for every condition, the



Fig. 6. Interface for Experiment III. To give the response, listeners clicked on one of the small blue rectangles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

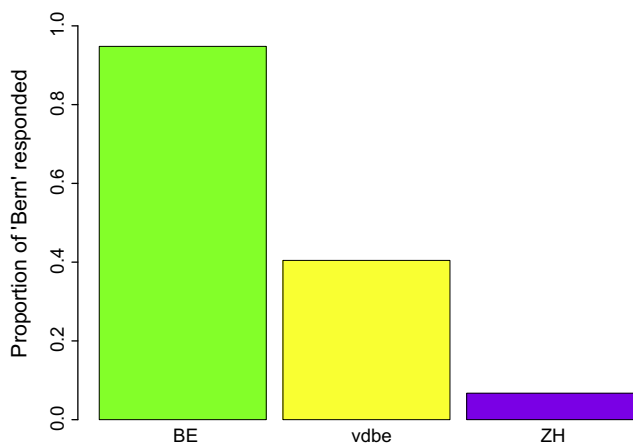


Fig. 7. Bar chart of responses by condition: Bern German speech (left bar, green), voice disguised speech (middle bar, yellow), and Zurich German speech (right bar, purple).

proportion of “Bern” responses. When presented Bern German speech, the Bern German listeners categorized it as speech from Bern German speakers 95% of the time (mean) – as indicated by the leftmost green column in Fig. 7. Zurich German was perceived as Bern German 7% of the time – as indicated by the rightmost purple bar in Fig. 7. Zurich German speakers imitating Bern German were perceived as Bern German speakers 40% of the time, indicated by the yellow bar in the middle. Inferential statistics show that the conditions significantly differed from one another ($p < .0001^*$, AIC = 4975).

Fig. 8 shows boxplots by speaker of the listeners’ responses only to the stimuli containing voice-disguised speech.

Three out of ten speakers were categorized as Bern German speakers above or right at chance level: speaker 14 m ($M = 70$, $SD = 24$), speaker 07 m ($M = 56$, $SD = 22$), and speaker 06f ($M = 50$, $SD = 24$). Speaker 12 m, on the other hand, was almost always rejected as Bern German speaker ($M = 11$, $SD = 8$). Differences between speakers’ Bern German imitation, as perceived by the listeners, were significant ($p < .0001^*$, AIC = 3439; Bonferroni-adjusted for condition, $\alpha = .016$). Using the R package *multcomp* (Hothorn et al., 2008), we observed that 29 of 45 cross-comparisons between speakers were significant (Bonferroni-adjusted for speaker, $\alpha = .005$; note that for multiple comparisons, speaker had to be entered as fixed effect into the model). Furthermore, listeners significantly differed from one another in their responses ($p < .0001^*$, AIC = 4975).

As for the confidence scores – where the higher the score, the less confident the listener – listeners were most certain in their responses when rating the Bern German stimuli ($M = 1.11$, $SD = .07$) and almost equally confident when rating the Zurich German stimuli ($M = 1.12$, $SD = .11$). When rating the disguised stimuli, listeners’ confidence levels decreased ($M = 1.38$, $SD = .16$).

In the following analyses we examined the relationship between the perception results from Experiment III and the production data in Experiment II. Do those three speakers that pass as Bern speakers above or at chance level in the

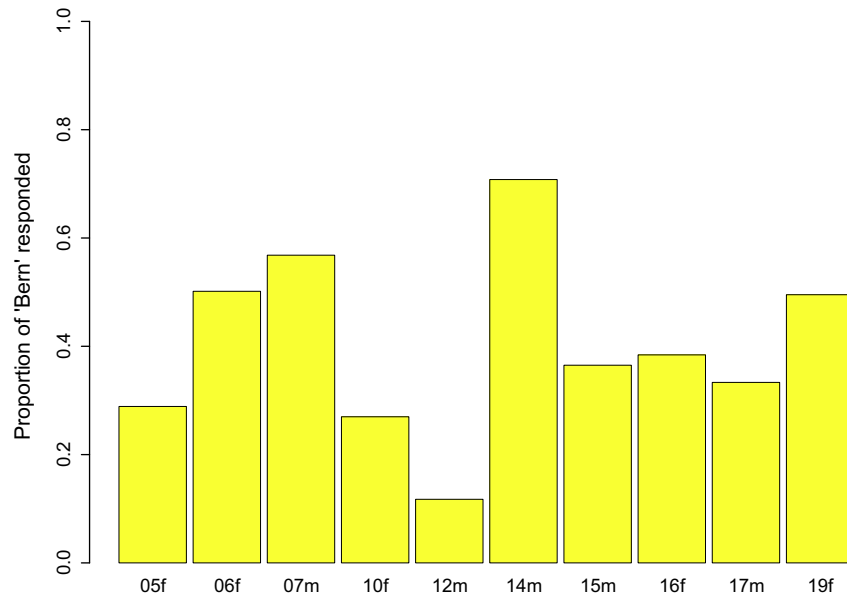


Fig. 8. Bar chart of the listeners' responses for the voice disguise condition, crossed by speaker.

perception test – speakers 06 f, 07 m and 14 m – differ in their production of suprasegmental temporal measures from those who were rejected as Bern German speakers? Based on the insights gained from Experiment III, we categorized these three speakers as 'good imitators' and the other seven as 'poor imitators' and tested for an effect of *performance*. We would assume good imitators to perform more closely to the target Bern German, i.e. to demonstrate lower suprasegmental temporal variability and lower percentages of voiced portions than in their native Zurich German (see Experiment I), than the poor imitators. To test for the effect of *performance*, *gender* and *performance* were included as fixed effects in a full linear mixed effects model, whereas *sentence* and *speaker* were treated as random effects (R code: `modelVD_full <- lmer(percentVoiced ~ gender + performance + (1|speaker) + (1|sentence), data = data)`). The reduced model for *performance* was coded as `modelVD_reduced <- lmer(percentVoiced ~ gender + (1|speaker) + (1|sentence), data = data)`. Fig. 9 shows the boxplots of the good imitators' (left) and poor imitators' (right) suprasegmental temporal measures.

The model with *performance* as a fixed effect provided an improved goodness of fit in none of the suprasegmental temporal measures. Descriptively, %VO shows the greatest difference between the two groups: those who performed better at imitating the Bern dialect show lower values ($M = 63.7\%$, $SD = 8.3$) than those that were poor at imitating ($M = 65.7\%$, $SD = 7.5$). In *varcoPeak*, and *nPVI_peak*, we find a similar trend in that good imitators revealed lower suprasegmental temporal variability values than poor imitators. In *varcoVO* and *nPVI_VO*, good imitators showed slightly higher values than poor imitators – again, this observation was statistically not significant.

5. Discussion

In the present paper we reported evidence that (a) the two dialects Bern and Zurich German differ from one another in *varcoPeak* (if we conservatively treat *speaker* as a random effect), (b) speakers seem to carry over some suprasegmental temporal patterns when disguising their voices (in particular *nPVI_VO*), and (c) speakers who were

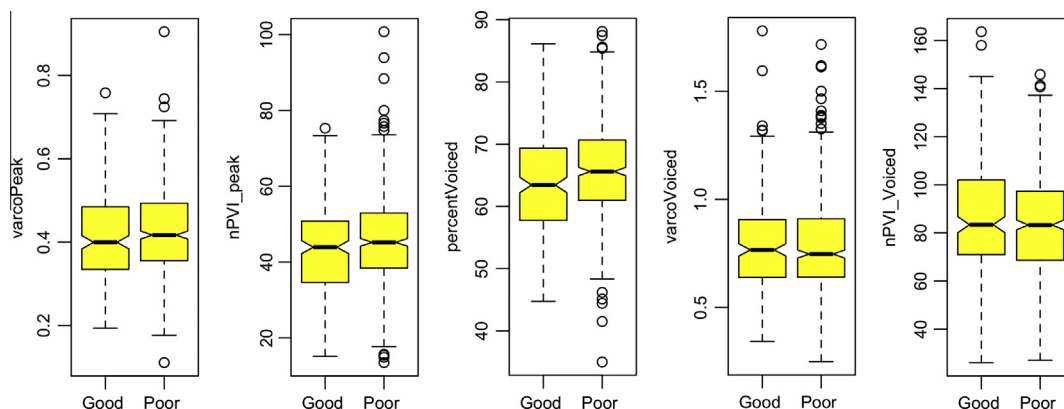


Fig. 9. Boxplots of the two speaker groups' suprasegmental temporal measures: good imitators (left) and poor imitators (right).

perceived as authentic in their imitations may have succeeded in doing so in part by lowering their global peak-to-peak variability and their amount of voicing towards that of Bern German, amongst other features. Here we discuss the results of the three experiments in detail and elaborate on potential implications for forensic voice comparison.

5.1. Experiment I

The results of the first experiment showed significant suprasegmental temporal differences between the two dialects in the durational variability of peak-to-peak intervals. It is possible that the smaller durational variability in Bern German could be explained by the afore-mentioned differences in morpho-phonology between the two dialects: processes such /nd/-velarization and /l/-vocalization both contribute to Bern German having less complex syllable structures (see Introduction). This may lead to smaller suprasegmental temporal variability: peak-to-peak interval durations are likely to become smaller when consonant clusters are reduced, i.e. more similar to the durations between peaks that are located in less complex syllables. As for the descriptively lower %VO in Bern German, we can only speculate. /nd/-velarization should not have an effect on %VO, as in Swiss German /d/ is voiceless to begin with (cf. Fleischer and Schmid, 2006). That is, voicing should not be affected by whether or not a Swiss German speaker says [hʊŋ] or [hʊŋd] *dog*. /l/-vocalization, should also not affect %VO as both /l/ and /ʊ/ are voiced. The finding of low %VO for Bern German is in line with Leemann et al. (2012), however, who reported low percentages over which speech is vocalic for Bern German (the two measures – %VO and the percentage over which speech is vocalic – covary, cf. Leemann et al., 2014a).

The purpose of having speakers imitate a foreign dialect was to introduce within-speaker variability. The results of Experiment I indicate that the dialects are significantly different in one of the applied measures, varcoPeak – descriptively in all of them, however. Further research is required to unpick possible reasons for between-dialect differences in voicing and peak-to-peak interval variability. Moreover, it remains to be tested whether or not these dialectal differences in suprasegmental temporal features are perceptually salient.

5.2. Experiment II

Results from the second experiment revealed speaker-specific suprasegmental temporal patterns which, in some instances, remain invariant even when the speakers imitate another dialect. This is particularly true for nPVI_VO, which exhibited high between-speaker variability and low within-speaker variability (cf. Table 2), which is desirable for forensic speaker comparison (Nolan, 2009). This is interesting given that previous research has shown nPVI_VO to be only moderately speaker-specific. Leemann et al. (2014a)

have shown that – of the automatic suprasegmental temporal measures examined – it was predominantly %VO and varcoPeak that carry speaker-specific information. Here we provide evidence that – with this corpus and the experimental design used – other mechanisms seem to be at work, resulting in nPVI_VO being highly speaker-specific. There are reports that suprasegmental temporal measures can be affected by a number of factors, e.g. speaking style (Leemann et al., 2014a), sentence material used (Wiget et al., 2010; Dellwo et al., 2015), as well as the labeler annotating the data (Wiget et al., 2010). This means that between-speaker differences found in one measure must not necessarily translate to similar between-speaker differences in the same measure when other sentence material is used, a different speaking style is elicited, or when the corpus was labeled by different researchers.

Results of Experiment II further revealed an effect of *disguise* in a number of measures, with the disguised sentences having significantly higher suprasegmental temporal variability than native Zurich German (varcoPeak, nPVI_peak, and %VO). Imitated Bern German, apparently, features higher peak-to-peak interval duration variability and a higher %VO than Zurich German. This finding is noteworthy, as results from Experiment I showed an opposite trend for authentic Bern German: Zurich German revealed higher values than authentic Bern German for all of the tested suprasegmental temporal measures. When imitating Bern dialect, then, the Zurich speakers perform in an opposite direction, generating variability in suprasegmental temporal values that are much higher than those of native Bern German speakers.

Qualitative analyses revealed segmental and suprasegmental features that may help explain this performance of the imitators. An informal check of a subset of sentences showed, for example, that in trying to imitate the typically slower articulation rate of Bern German, Zurich German speakers stretch – open syllables in particular – exceptionally much. On the other hand they truncate other syllables to an unnatural extent in the imitation task. This may contribute to higher variability particularly in varcoPeak and nPVI_peak. Furthermore, stretching open syllables may result in stretching mainly the vowel, leading to a longer voiced portion, i.e. a higher %VO. In a foreign accent imitation task, Neuhauser (2011a) found that imitators tend to unnaturally exaggerate certain phonetic features of a foreign accent – which seems to apply to a dialect imitation task as well, as shown by the present findings. More generally, research in second language acquisition has shown, too, that learners of a foreign language have difficulties in acquiring target language durational features – such as for example the percentage over which speech is vocalic (White and Mattys, 2007). Second language speakers have been shown to lengthen vowels (Adams and Munro, 1978; Taylor, 1981). Perhaps, speakers use similar strategies in dialect imitation and foreign language learning.

Further impressionistic examination revealed that two speakers exhibited less laryngalization in the imitated Bern

German as opposed to their natural Zurich German speech. This translates to higher portions of automatically detected periodic parts in the signal, i.e. an increase in %VO in the disguise condition. Moreover, some speakers – while probably intending to capture the slowness of Bern German – speak somewhat less enunciated and (uncommon for Swiss German) articulate some /s/ as /z/. This again may boost %VO values. The good imitators that we impressionistically analyzed did not seem to follow these patterns, i.e. were less likely to lengthen open syllables to such an unusual degree, and less likely to truncate certain syllables to the degree of the poor imitators. Moreover, none of them seemed to change their voice quality during imitation. As for varcoVO, the boxplot in Fig. 4 showed somewhat higher variability in this parameter for the disguise condition. Here too, this may have to do with the unusual lengthening and truncation of some syllables. What is interesting, though, is that this truncation and lengthening may be occurring more often in non-adjacent segments, as the pairwise variability index of the voiced interval durations (nPVI_VO) in the signal seems to be less affected by this variability.

In forensic voice comparison (FVC), experts typically compare a number of speaker-individual features between trace and comparison material. The most common parameters currently being examined in FVC are voice fundamental frequency and vocal tract resonance frequencies (Nolan, 2001; Gold and French, 2011). In addition, acquired features such as dialect, accent, or sociolects are examined. In FVC it is critical to collect as many features of the speech signal as possible (French et al., 2010). With the research conducted here, we report further characteristics in the suprasegmental temporal domain that seem to remain invariant, even when imitating a foreign dialect. Such suprasegmental features seem to be more robust than other acoustic parameters that were hitherto examined: for voice disguised speech, previous studies reported a change in long-term spectra (Hollien and Majewski, 1977), spectral characteristics of fricatives (Fecher and Watt, 2011; Hirson and Duckworth, 1993), f0 and formant frequencies (Eriksson and Wretling, 1997). Clearly, all of these studies applied different techniques of voice disguise, which obstructs direct comparisons between their results. Nevertheless it is noteworthy that one of the measures applied here, nPVI_VO, seems to withstand within-speaker variability as introduced by imitating a foreign dialect. An integration of such features in FVC work may thus enhance and complement the search for speaker-specificity in trace and suspect material – in those cases where speakers change their regional accent as a technique for voice disguise.

5.3. Experiment III

Experiment III showed that the Bern listeners performed well in detecting native Bern German speakers. At the same time they correctly rejected native Zurich German speakers

as Bern German speakers 93% of the time. As for the disguised condition, the Zurich imitators were accepted as Bern speakers only 40% of the time, i.e. 10% below chance threshold (since a two-alternative forced choice entails a 50% chance level). Native Bern German listeners seem to be very sensitive to imitated variants of their own variety. They seem to have a very clear idea of what authentic Bern German sounds like. In foreign accent imitation tasks however, Neuhauser (2011a) found that native listeners perform poorly at distinguishing imitated foreign accents from real foreign accents. There may be two explanations for this difference in listener performance. Firstly, the expected – and thus accepted – between-speaker variability in a foreign-accented ‘variety’ (e.g. French-accented German, Neuhauser, 2011a) may be higher than in a dialectal variety. While a particular dialectal variety is defined by a rather discrete and stable set of phonetic features, speakers of a particular foreign-accented ‘variety’ differ in the number and in the particular set of target-deviant features they use on the one hand, and also in the magnitude of this deviance on the other. This may result in some speakers being perceived as having a stronger accent than others (Cunningham-Andersson and Engstrand, 1989; Neuhauser, 2011b). Secondly, the task for listeners who judge imitators of their own dialect is to assess whether the speaker is ‘one of them’, whereas native listeners who take decisions about foreign-accented speech decide whether the speaker is part of a group that is not their own. This latter case may be a more demanding task: in the former case, the listeners function as ‘gatekeepers’ of their own variety, in the latter they do not (Nolan, 2012).

This finding may have relevance for forensic phonetics: earwitnesses may thus perform relatively well at detecting whether or not a criminal was speaking the same dialect as the earwitness or whether it was an imitation. At the same time, the high rate of rejections of imitations as authentic speech indicates a high sensitivity and dialectal awareness on the part of the listeners.

Results further revealed that good imitators behave differently in production compared to those who perform poorly in the dialect imitation task. In three out of five temporal measures, the good imitators performed more like native Bern speakers than the poor imitators: their varco-Peak, nPVI_peak, and %VO values were descriptively lower in their voice-disguised Bern dialect than in their native Zurich dialect, which corresponds to the pattern found in the native Bern and Zurich speakers in Experiment I. Perhaps some of the good imitators’ suprasegmental temporal features may be of perceptual relevance: Zurich German speakers with measurable more “Bern-like” suprasegmental temporal features were more often perceived as authentic speakers of Bern German.

Further seeking for more global durational differences between good and poor imitators, we performed automatic analyses of the good and poor performers’ speaking rates. To this end, we measured the number of automatically detected amplitude peaks per second with the same script

that performs peak detection described in Section 2.4. Based on the 74 sentences per speaker from Experiment II, results showed that good performers overall tend to speak faster ($M = 5.6$ peaks/s., $SD = .78$) than poor performers ($M = 5.3$ peaks/s., $SD = .78$); however, the effect of *performance* was not significant in an LME carried out as described in Section 4. Using a less conservative approach by defining *speaker* as a fixed instead of a random effect, the effect of *performance* becomes highly significant ($p < 0.0001$, $AIC = 1584$). Munro and Derwing (2001) have shown a relationship between accentedness in second language speech and speaking rate, where particularly slow or particularly fast speaking rate was associated with higher accentedness. The eight Bern German speakers from Experiment I on average demonstrated 5.8 peaks/s. Potentially, then, passing as an authentic Bern German imitator may also particularly involve capturing the articulation rate of the dialect – given that good performers of Experiment III perform more closely to actual Bern German with the 5.6 peaks/s. The poor performers may have been perceived as caricatures of Bern German speaking rate, having exaggerated a particularly slow articulation rate. Perhaps this is another reason why they were rejected as authentic Bern German speakers.

In the present study we examined read speech to study within-speaker variability in disguised and non-disguised speech. We are aware that in forensic casework, spontaneous speech is the more common form: criminal recordings are often spontaneous, while suspect recordings can be read and spontaneous. At the German Federal Criminal Police Office (BKA), for instance, an estimated 10–20% of criminal recordings is read. Read suspect material, on the other hand, can make up nearly 30% (Olaf Köster, personal communication). The majority of both criminal and suspect material is spontaneous. We decided to begin testing our assumptions on controlled read speech, given that (a) this would allow for a direct comparison between disguised and non-disguised speech and (b) previous research (Wiget et al., 2010; Dellwo et al., 2015) has shown strong effects of sentence material on suprasegmental temporal measures. We needed a tightly controlled corpus as differences in sentence material would have confounded our analyses of suprasegmental temporal measures. In future studies, the same hypotheses should be tested on a large corpus (so as to level out sentence effects) of forensically realistic spontaneous speech, to validate or question the findings put forth in the present study.

6. Conclusion

In this paper we explored the effect of a voice disguise technique – the imitation of another dialect – on suprasegmental temporal features. In some of the examined features, results revealed high-between speaker variability and, at the same time, low within-speaker variability across

the disguise condition. In the rate-normalized average differences between consecutive voiced interval durations (nPVI_VO), we find speaker-individuality that remains fully invariant, regardless of whether the speech is disguised or not. Such knowledge may complement our knowledge of speaker-invariant characteristics, such as found in the frequency domain (formant frequencies, e.g. McDougall, 2004, 2006; fundamental frequency, e.g. Künzel, 2000) or in the intensity domain (Amino and Arai, 2009). Our findings further indicated that imitating another dialect and passing as a native speaker of that dialect is a highly challenging task. Yet, those speakers who performed well at imitating the dialect may have passed as native speakers of that dialect possibly also because they were able to approximate suprasegmental temporal features (varcoPeak and %VO values in particular) of the target dialect, amongst many other segmental and suprasegmental features. The current findings and their potential implications are based on a highly controlled set of data. In the future, analyses on spontaneous data will have to be carried out to test the present hypotheses.

Acknowledgements

This research is supported by the Swiss National Science Foundation (SNSF), grant no. 100015_135287. We would like to thank three anonymous reviewers and Paul Foulkes for very helpful comments on earlier versions of this manuscript. The authors would like to further thank Volker Dellwo, Stephan Schmid, Ingrid Hove, Dario Brander, and Sibylle Sutter for their contribution to these corpora and discussions leading up to this paper.

Appendix A

- 1
BE: Dr Gloon hat es luschtegs Gsecht gha.
ZH: Dä Gloon hät äs luschtigs Gsicht gha.
EN: The clown had a funny face.
- 2
BE: Dr Motor louft.
ZH: Dä Motor läuft.
EN: The car engine's running.
- 3
BE: Si hät mit em Mässer gschnettä.
ZH: Si hät mit em Mässer gschnittä.
EN: She cut with her knife.
- 4
BE: D Ching hei gärn Erbbeerli.
ZH: D Chind händ gern Erbberli.
EN: Children like strawberries.
- 5
BE: S Hus hät nün Zemmr.
ZH: S Huus hät nün Zimmer.
EN: The house had nine rooms.

- 6
BE: Si choufe Brot.
ZH: Si chaufed Brot.
EN: They're buying some bread.
- 7
BE: Di grüäne Tomate si no chli.
ZH: Di grüänä Tomatä sind no chli.
EN: The green tomatoes are small.
- 8
BE: Sie luege uf d Uhr.
ZH: Si lueged uf d Uhr.
EN: They're looking at the clock.
- 9
BE: Er hät mit sim Zog gspiult.
ZH: Er hät mit sim Zug gschpilt.
EN: He played with his train.
- 10
BE: Dä Pöschtlter häts Gartetörli zuegmacht.
ZH: Dä Pöschtlter hät s Gartätöörli zuägmacht.
EN: The postman shut the gate.
- 11
BE: Dä Sack isch ufe Bode keit.
ZH: Dä Sack isch uf dä Bodä gheit.
EN: The bag bumps on the ground.
- 12
BE: Dr Bueb hät e Hangstand gmacht.
ZH: Dä Buüb hät en Handstand gmacht.
EN: The boy did a handstand.
- 13
BE: E Chatz hock ufem Bett.
ZH: A Chatz hockt uf em Bett.
EN: A cat sits on the bed.
- 14
BE: Dr Laschtwage hät Frücht gliferet.
ZH: Dä Laschtwagä hät Frücht gliferet.
EN: The lorry carried fruit.
- 15
BE: Es het i Ströme grägnät.
ZH: Es hät in Ströhmä grägnät.
EN: The rain came down.
- 16
BE: S Glacé isch roserot gsi.
ZH: S Glacé isch rosärot gsi.
EN: The ice cream was pink.
- 17
BE: Dr Chessu steit bi de Törä.
ZH: De Chessel staht bi dä Türä.
EN: The ladder's near the door.
- 18
BE: Si hei e schöne Tag gha.
ZH: Si händ en schönä Tag gha.
EN: They had a lovely day.
- 19
BE: Dr Bau isch is Gool gangä.
ZH: Dä Ball isch is Gool ggängä
- EN: The ball went into the goal.
- 20
BE: Die aute Häntschä si dräckig.
ZH: Di altä Händschä sind dräckig.
EN: The old gloves are dirty.
- 21
BE: Er het sich ine Finger gschnitte.
ZH: Er hät sich in en Finger gschnittä.
EN: He cut his finger.
- 22
BE: Dr Schäferhung hät Hunger gha.
ZH: Dä Schäferhund hät Hunger gha.
EN: The thin dog was hungry.
- 23
BE: Dr Bueb häts Schpel kännt.
ZH: Dä Buüb hät s Spiel kännt.
EN: The boy knew the game.
- 24
BE: Es schneit a Heiligabe.
ZH: Es schneit an Heiligobe.
EN: Snow falls at Christmas.
- 25
BE: Si nimmt ihre Mantu.
ZH: Si nimmt ihrän Mantel.
EN: She's taking her coat.
- 26
BE: D Polizei vefolgt es Outo.
ZH: D Polizei verfolgt äs Auto.
EN: The police chased the car.
- 27
BE: S Müsli isch ines Loch krochä.
ZH: S Müsli isch inäs Loch krochä.
EN: A mouse ran down the hole.
- 28
BE: D Frou baschtlet es Speuzüg.
ZH: D Frau baschtläd es Spielzüg.
EN: The lady's making a toy.
- 29
BE: Es lege es paar Frücht ungerem Boum.
ZH: Es ligged es paar Frücht unter em Baum.
EN: Some sticks were under the tree.
- 30
BE: S chline Bebe schlaft.
ZH: S chliine Baby schlaft.
EN: The little baby sleeps.
- 31
BE: Sie luege em Zug zue.
ZH: Si lueged am Zug zuä.
EN: They're watching the train.
- 32
BE: Hüt het d Schuäu früecher agfange.
ZH: Hüt hät d Schuäl früäner aagfange.
EN: The school finished early.

33

BE: D Glasschösslä isch kabott gange.

ZH: D Glasschüsslä isch kaputt gange.

EN: The glass bowl broke.

34

BE: Dr Hung hät met em Schpiuzüg gschpiut.

ZH: Dä Hund hät mit em Spiilzüg gspilt.

EN: The dog played with a stick.

35

BE: Dr Wasserchocher esch rächt heiss.

ZH: Dä Wasserchocher isch rächt heiss.

EN: The kettle's quite hot.

36

BE: Dr Puur hät e Chue.

ZH: Dä Puur hät e Chue.

EN: The farmer keeps a bull.

37

BE: Si diskutiäre läbr langsam.

ZH: Si diskutieräd läber langsam.

EN: They say some silly things.

38

BE: D Frou hät dä Mantu agha.

ZH: D Frau hät en Mantel aagha.

EN: The lady wore a coat.

39

BE: D Ching si wegglofe.

ZH: D Chind sind weg ggloffä.

EN: The children are walking home.

40

BE: Är hät dringend Ferie brucht.

ZH: Er hät dringend Feriä bruucht.

EN: He needed his holiday.

41

BE: D Miuch isch innärä Fläsche gse.

ZH: D Milch isch innärä Fläschä gsi.

EN: The milk came in a bottle.

42

BE: Dr Ma hät sini Schue potzt.

ZH: Dä Maa hät sini Schuä putzt.

EN: The man cleaned his shoes.

43

BE: Sie hei Vannilglacé gässe.

ZH: Si händ Vanilleglacé ggässä.

EN: They ate the lemon jelly.

44

BE: Dr Bueb esch wäggloffä.

ZH: Dä Buäb isch wägg gloffä.

EN: The boy's running away.

45

BE: Dr Vater hät sech s Buech agluegt.

ZH: Dä Vater hät sich s Buäch aagluegt.

EN: Father looked at the book.

46

BE: Sie trinkt us nöiä Tasse.

ZH: Si trinkt us neuä Tassä.

EN: She drinks from her cup.

47

BE: S Zemmer wird chaut.

ZH: S Zimmer wird chalt.

EN: The room's getting cold.

48

BE: S Meitli (Meitschi) isch zom Tisch gloufe.

ZH: S Meitli isch zum Tisch gloffä.

EN: A girl kicked the table.

49

BE: D Frou hät ehrem Ma ghoufe.

ZH: D Frau hät ihrem Maa gholffä.

EN: The wife helped her husband.

50

BE: D Maschenä hät feu Lärm gmacht.

ZH: D Maschinä hät vil Lärm gmacht.

EN: The machine was quite noisy.

51

BE: Dr auti Maa macht sech Sorgä.

ZH: Dä alti Maa macht sich Sorge.

EN: The old man worries.

52

BE: Ä Bueb isch dr Wäg zrügg gloffä.

ZH: Än Buäb isch dä Wäg zrug ggloffä.

EN: A boy ran down the path.

53

BE: S Hus hät e schönä Garte gha.

ZH: S Huus hät en schönä Gartä gha.

EN: The house had a nice garden.

54

BE: Si het mit ehrem Sohn grett.

ZH: Si hät mit ihrem Sohn gredd.

EN: She spoke to her son.

55

BE: Si loufe öber d Schtrass.

ZH: Si laufed über d Straass.

EN: They're crossing the street.

56

BE: Zitrone wachse uf de Böim.

ZH: Zitronä wachsed uf dä Boim.

EN: Lemons grow on trees.

57

BE: Er hät si Brüeder gfunge.

ZH: Er hät sin Brüäder gfundä.

EN: He found his brother.

58

BE: Es paar Tier schlafe ufem Stroh.

ZH: Es paar Tier schlaafed ufm Stroh.

EN: Some animals sleep on straw.

- 59
BE: Ds Gonfiglas isch vou gsi.
ZH: S Comfi-Glas isch voll xi.
EN: The jam jar was full.
- 60
BE: Sie lege ufem Bode.
ZH: Si ligged uf em Bodä.
EN: They're kneeling down.
- 61
BE: Se Meitli hät ehres Bebe vrlöre.
ZH: S Meitli hät ihres Baby verlohrä.
EN: The girl lost her doll.
- 62
BE: Dr Choch bacht e Chueche.
ZH: Dä Choch bacht en Chuächä.
EN: The cook's making a cake.
- 63
BE: S Ching nemt e Schpiuzüg id Hang.
ZH: S Chind nimmt es Spielzüg id Hand.
EN: The child grabs the toy.
- 64
BE: A sine Schue häts Dreck gha.
ZH: A sinä Schuä häts Dräck gha.
EN: The mud stuck on his shoe.
- 65
BE: S Badtuech esch nass gsi.
ZH: S Badtuäch isch nass gsi.
EN: The bath towel was wet.
- 66
BE: D Zündhöuzli si ufem Gschteu gläge.
ZH: D Zündhölzli sind uf em Gstell glägä.
EN: The matches lie on the shelf.
- 67
BE: Si si am Hus vrbi gloufe.
ZH: Si sind am Huus verbi ggloffe.
EN: They're running past the house.
- 68
BE: Dr Zog hät ä schwäre Unfau gha.
ZH: Dä Zug hät en schwäre Unfall gha.
EN: The train had a bad crash.
- 69
BE: D Badwannä isch läär.
ZH: D Badwanne isch läär.
EN: The kitchen sink's empty.
- 70
BE: E Bueb isch usem Fenschter gschtegä.
ZH: En Bueb isch us em Fänschter gschtige.
EN: A boy fell from the window.
- 71
BE: Si hät ihrä Löffu brucht.
ZH: Si hät ihren Löffel bbruchet.
EN: She used her spoon.
- 72
BE: Dr Park let gad be dä Strass.
ZH: Dä Park lyt grad bi dä Straass.
EN: The park's near the road.
- 73
BE: Dr Choch hät Zebälä gschnettä.
ZH: Dä Choch hät Zwiblä gschnüttä.
EN: The cook cut some onions.
- 74
BE: Dr Hung hät lisli knurrt.
ZH: Dä Hund hät liislig knurred.
EN: The dog made an angry noise.
- 75
BE: Er wäscht ses Gsicht.
ZH: Er wäscht sis Gsicht.
EN: He's washing his face.
- 76
BE: Öpper häts Gäud gno.
ZH: Öppert hät s Gält gno.
EN: Somebody took the money.
- 77
BE: S Liecht esch usgangä.
ZH: S Liecht isch uusggangä.
EN: The light went out.
- 78
BE: Si hei es paar Härdöpfu wöuä.
ZH: Si händ es paar Herdöpfel welä.
EN: They wanted some potatoes.
- 79
BE: S Luusmeitli schreit umenang.
ZH: S Luusmeitli schreit umänand.
EN: The naughty girl's shouting.
- 80
BE: Di chaut Miuch isch em Chruäg.
ZH: Di chalt Milch isch im Chruäg.
EN: The cold milk's in a jug.
- 81
BE: D Farb esch ufe Bode tropft.
ZH: D Farb isch uf dä Bodä tropft.
EN: The paint dripped on the ground.
- 82
BE: D Mueter hät de Tee ufgosse.
ZH: D Mueter hät dä Tee ufgosse.
EN: The mother stirs the tea.
- 83
BE: Si hei über d Gschecht glacht.
ZH: Si händ über d Gschicht glacht.
EN: They laughed at his story.
- 84
BE: Männer trage langi Hosä.
ZH: Männer träged langi Hosä.
EN: Men wear long trousers.
- 85
BE: Dr Buäb hät gschlaffe.
ZH: Dä Bueb hät gschlafä.
EN: The small boy was asleep.
- 86
BE: D Sunnä hät de Schnee gschmouze.
ZH: D Sunä hät dä Schnee gschmolzä.
EN: The sun melted the snow.

87

BE: Dr Fatter chunt hei.

ZH: Dä Vater chunnt hei.

EN: The father's coming home.

88

BE: Si hät ehres erschti Feriegäud gha.

ZH: Si hät ihres erschtä Feriegäld gha.

EN: She had her pocket money.

89

BE: Dr Laschtwage esch uf dr Schtraas gfahre.

ZH: Dä Laschtwagä isch uf de Straass gfahrä.

EN: The lorry drove up the road.

90

BE: Är bringt si Rägämantu met.

ZH: Er bringt sin Rägämantel mit.

EN: He's bringing his raincoat.

91

BE: D Frou geit zom Ladä.

ZH: D Frau gaht zum Ladä.

EN: The lady goes to the shop.

92

BE: Es scharfs Mässer esch gfärlech.

ZH: Äs scharfs Mässer isch gföhrlich.

EN: A sharp knife's dangerous.

93

BE: Si hei öppis zom Zmettag metgno.

ZH: Si händ öppis zum Zmittag mitgnoh.

EN: They took some food.

94

BE: Di gschide Meitli se am Läse.

ZH: Di gschydä Meitli sind am Läsä.

EN: The clever girls are reading.

95

BE: Dr Bäse esch em Ägge gschtange.

ZH: Dä Bäsä isch im Eggä gstandä.

EN: The broom stood in the corner.

96

BE: D Frou hät ehres Huus putzt.

ZH: D Frau hät ihres Huus putzt.

EN: The woman tidied her house.

97

BE: D Ching hei d Täsche lege lo.

ZH: D chind händ d Täschä ligge lah.

EN: The children dropped the bag.

98

BE: Dr Hung esch zrüggho.

ZH: Dä Hund isch zrugg cho.

EN: The dog came back.

99

BE: Dr Bode hät suber usgseh.

ZH: Dä Bodä hät suuber usgseh.

EN: The floor looked clean.

100

BE: Si hät ehri Hangtäsche gfunge.

ZH: Si hät ihri Handtäschä gfundä.

EN: She found her purse.

101

BE: D Frucht legt am Bode.

ZH: D Frucht lyt am Bodä.

EN: The fruit lies on the ground.

102

BE: D Mueter nemt e Pfanne.

ZH: D Mueter nimmt ä Pfanä.

EN: Mother fetches a saucepan.

103

BE: Si hei mit chautem Wassr gwäsche.

ZH: Si händ mit chalem Wasser gwäschä.

EN: They washed in cold water.

104

BE: Di jonge Lüt sei am Tanzä.

ZH: Di jungä Lüüt sind am Tanzä.

EN: The young people are dancing.

105

BE: Dr Bos esch z früe cho.

ZH: Dä Bus isch zfrüäh cho.

EN: The bus went early.

106

BE: Sie hei zwöi läri Fläsche gha.

ZH: Si händ zwei lääri Fläschä gha.

EN: They had two empty bottles.

107

BE: E Fuesbau rollt es Fäud.

ZH: En Fuessball rollt is Fäld.

EN: A ball's bouncing along.

108

BE: Dr Vater häts Brot vrgässe.

ZH: Dä Vatter hät s Brot vergässä.

EN: The father forgot the bread.

109

BE: D Mueter hät es Biuderbuäch.

ZH: D Muetr hät äs Bilderbuäch.

EN: The girl has a picture book.

110

BE: D Orangsche esch zimlech süäss gse.

ZH: D Orange isch ziemlich süäss gsi.

EN: The orange was quite sweet.

111

BE: Er häbt d Nase zue.

ZH: Er hebt d Nasä zuä.

EN: He's holding his nose.

112

BE: D Straass esch uf de Charte.

ZH: D Strass isch uf dä Chartä.

EN: The new road's on the map.

113

BE: Dr Bueb het ses Buech vrgässe.

ZH: Dä Buäb hät sis Buäch vergässä.

EN: The boy forgot his book.

114

BE: Dr Kolleg esch zom Zmettag cho.

ZH: Än Kolleg isch zum Zmittag cho.

EN: A friend came for lunch.

115

BE: Dr Wärbzügchaschte esch lär.
 ZH: De Wärbzügchaschte isch läär.
 EN: The match boxes are empty.

116

BE: Är esch ufe Wage gschtege.
 ZH: Er isch uf de Wagä gschtinge.
 EN: He climbed his ladder.

117

BE: Dr Chruog esch ufem Gschteu gschtange.
 ZH: Dä Chruäg isch uf em Gstell gstandä.
 EN: The jug stood on the shelf.

118

BE: Dr Fuesbau hät s Fenschter kabott gmacht.
 ZH: Dä Fuessball hät s Fänschter kaputt gmacht.
 EN: The ball broke the window.

119

BE: Si choufe Chäs.
 ZH: Si chaufed Chäs.
 EN: They're shopping for cheese.

120

BE: D Familie hät es Huus kouft.
 ZH: D Familiä hät es Huus gkauft.
 EN: The family bought a house.

121

BE: S Wasser vom Weier esch dräckege.
 ZH: S Wasser vom Weier isch dräckig.
 EN: The pond water's dirty.

122

BE: D Polezei potzt d Schtraas.
 ZH: D Polizei putzt d Strass.
 EN: Police are clearing the road.

123

BE: Dr Bos hät plötzlech brämst.
 ZH: Dä Bus hät plötzlich brämst.
 EN: The bus stopped suddenly.

124

BE: Si schribt erem Brüeder.
 ZH: Si schrybt ihrem Brüäder.
 EN: She writes to her brother.

125

BE: Dr Fuesbauer hät e Schue välore.
 ZH: Dä Fuessballer hät en Schuä verlorä.
 EN: The footballer lost a boot.

126

BE: Di drüü Meitli lose.
 ZH: Di drüü Meitli losed.
 EN: The three girls are listening.

127

BE: Dr Mantu ligt ufm Stouou.
 ZH: Dä Mantel lyt uf em Stuäl.
 EN: The coat lies on a chair.

128

BE: I däm Buoch wird ä Gschicht vezeut.
 ZH: I dem Buäch wird ä Gschicht verzellt.
 EN: The book tells a story.

129

BE: Dr Bueb esch zrügge gange.
 ZH: Dä Buüb isch zrug ggängä.
 EN: The young boy left home.

130

BE: Si chlättere ufe Boum.
 ZH: Si chlättäräd uf en Baum.
 EN: They're climbing the tree.

131

BE: Si esch näbeme Fänschter gschtange.
 ZH: Si isch näbämä Fänschter gstandä.
 EN: She stood near her window.

132

BE: Dr Tesch hät drü Bei.
 ZH: Dä Tisch hät drüü Bei.
 EN: The table has three legs.

133

BE: E Brief esch ufe Bode keit.
 ZH: Än Brief isch uf dä Bodä gheit.
 EN: A letter fell on the mat.

134

BE: Fünf Männer se am schaffe.
 ZH: Foif Männer sind am schaffä.
 EN: The five men are working.

135

BE: D Schuä se sähr dräckege gse.
 ZH: D Schuä sind sehr dräckig gsi.
 EN: The shoes were very dirty.

136

BE: Er machts für si Vater.
 ZH: Er machts für sin Vatter.
 EN: He listens to his father.

137

BE: Si si id Ferie gange.
 ZH: Si sind id Feriä ggängä.
 EN: They went on holiday.

138

BE: S Bebe hät e Tasse kabottgmacht.
 ZH: S Baby hät ä Tassä kaputt gmacht.
 EN: Baby broke his mug.

139

BE: D Frou hät ehre Kofer packt.
 ZH: D Frau hät ihrä Koffer packt.
 EN: The lady packed her bag.

140

BE: Dr Täuer esch heiss.
 ZH: Dä Täller isch heiss.
 EN: The dinner plate's hot.

141

BE: Dr Zug fahrt schnäu.
 ZH: Dä Zug fahrt schnäll.
 EN: The train's moving fast.

142

BE: S Ching hät Meuch tronke.
 ZH: S Chind hät Milch trunkä.
 EN: The child drank some milk.

- 143
BE: S Outo hät e Wang gschtreift.
ZH: S Auto hät ä Wand gstreift.
EN: The car hit a wall.
- 144
BE: S Potzmittu esch ufem Küouschrank.
ZH: S Putzmittl isch uf em Chüelschrank.
EN: A tea towel's by the sink.
- 145
BE: D Potzfrou hät e Bäse brucht.
ZH: D Putzfrau hät en Bäsä brucht.
EN: The cleaner used a broom.
- 146
BE: Si hät en Spiägu gluogt.
ZH: Si hät in Spiegel gluegt.
EN: She looked in her mirror.
- 147
BE: Dr bravi Buob helft met.
ZH: Dä bravi Buüb hilft mit.
EN: The good boy's helping.
- 148
BE: Si si dr Wäg zrugg gloffä.
ZH: Si sind dä Wäg zrug gloffä.
EN: They followed the path.
- 149
BE: D Chochiuhr esch faosch gange.
ZH: D Chuchiuhr isch falsch ggangä.
EN: The kitchen clock was wrong.
- 150
BE: Öppr louft übr d Stross.
ZH: Öppert lauft über d Straass.
EN: Someone's crossing the road.
- 151
BE: Dr Böschtlr brengt ä Briäf.
ZH: Dä Pöschtlr bringt en Brief.
EN: The postman brings a letter.
- 152
BE: Dr Hung esch ufe Tesch gstegä.
ZH: Dä Hund isch uf dä Tisch gschtige.
EN: The dog jumped on the chair.
- 153
BE: Si si am Velofaare.
ZH: Si sind am Velofahrä.
EN: They're cycling along.
- 154
BE: Er het ses Bei brochä.
ZH: Er hät sis Bai bbrochä.
EN: He broke his leg.
- 155
BE: D Meuch esch vorem Hus gschlange.
ZH: D Milch isch vor äm Huus gstandä.
EN: The milk was by the front door.
- 156
BE: D Jacke hänge em Chaschte.
ZH: D Jagge hanged im Chaschte.
EN: The shirts hang in the cupboard.
- 157
BE: Dr Bode esch z härt gse.
ZH: Dä Bodä isch zhärt gsi.
EN: The ground was too hard.
- 158
BE: Dr Topf esch dicht.
ZH: Dä Topf isch dicht.
EN: The buckets hold water.
- 159
BE: S Huen hät es paar Eier gleit.
ZH: S Huän hät es paar Eier gleit.
EN: The chicken laid some eggs.
- 160
BE: D Familie chocht Fesch.
ZH: D Familiä chocht Fisch.
EN: The family like fish.
- 161
BE: Zucker isch sehr süess.
ZH: Zucker isch sehr süäss.
EN: Sugar's very sweet.
- 162
BE: S Bebe esch ufem Tüechli gläge.
ZH: S Baby isch ufäm Tüächli gglägä.
EN: The baby lay on a rug.
- 163
BE: D Wöschmaschine esch chabott gange.
ZH: D Wäschmaschinä isch kaputt ggangä.
EN: The washing machine broke.
- 164
BE: Si putze dä Tesch.
ZH: Si butzed dä Tisch.
EN: They're clearing the table.
- 165
BE: D Putzfrou hät dä Bode potzt.
ZH: D Putzfrau hät dä Bodä putzt.
EN: The cleaner swept the floor.
- 166
BE: Dr Vechöifer vechouft Bluomä.
ZH: Dä Verchoifer verchauft Bluämä.
EN: A grocer sells butter.
- 167
BE: Dr Chäser fahrt e chline Lifr wagä.
ZH: Dä Chäser fahr en chlinä Lifer wagä.
EN: The milkman drives a small van.
- 168
BE: S Badwasser esch warm gsi.
ZH: S Badewasser isch warm gsi.
EN: The bath water was warm.
- 169
BE: Är nimmt ä Löffu.
ZH: Er nimmt än Löffel.
EN: He's reaching for his spoon.
- 170
BE: Si hät sech ade Hang vrletzt.
ZH: Si hät sich a dä Hand verletzt.
EN: She hurt her hand.

171
BE: Dr Bueb esch uf dä Schtäge usgrütscht.
ZH: Dä Buüb isch uf dä Stägä uusgrutscht.
EN: The boy slipped on the stairs.

172
BE: Si blibe zum Znacht.
ZH: Si blybäd zum Znacht.
EN: They're staying for supper.

173
BE: S Meitli hät ä Spiegu ghebt.
ZH: S Meitli hät en Spiegel ghebt.
EN: The girl held a mirror.

174
BE: D Tasse esch ufem Ungertäur gschtange.
ZH: D Tassä isch ufämä Untertäller gstandä.
EN: The cup stood on a saucer.

175
BE: D Chüä se uf dä Markt bracht wordä.
ZH: D Chüä sind uf dä Markt bbraacht wordä.
EN: The cows went to market.

176
BE: Dr Bueb hät Schwierigkeite gärn gha.
ZH: Dä Bueb hät Schwirigkeitä gern gha.
EN: The boy got into trouble.

177
BE: Si göi i Usgang.
ZH: Si gönd in Uusgang.
EN: They're going out.

178
BE: Dr Fuäsbau hät de Pfoschte troffä.
ZH: Dä Fuässball hät dä Pfoschtä troffä.
EN: The football hit the goalpost.

179
BE: Er het d Rächnig zaut.
ZH: Er hät d Rächnig zallt.
EN: He paid his bill.

180
BE: Dr Pulli esch zemlech nass.
ZH: Dr Pulli isch zemlich nass.
EN: The teacloth's quite wet.

181
BE: E Chatz esch vom Wage gschtegä.
ZH: Ä Chatz isch vom Wage gschtige.
EN: A cat jumped off the fence.

182
BE: S Bebe hät bloui Ouge gha.
ZH: S Baby hät blauu Augä gha.
EN: The baby has blue eyes.

183
BE: D Mueter hät Vorhäng gnäit.
ZH: D Muäter hät Vorhäng gnäht.
EN: Mother made some curtains.

184
BE: Si si a dem Tisch gsässe.
ZH: Si sind a dem Tisch gsässä.
EN: They sat on a wooden bench.

185
BE: Dr Ofe esch z heiss.
ZH: Dä Ofä isch z heiss.
EN: The oven's too hot.

186
BE: S Meitli hät ke Fiebr meh.
ZH: S Meitli hät ke Fieber meh.
EN: The girl caught a cold.

187
BE: D Wentrjaggä esch ufghängt.
ZH: D Winterjagge isch uufghänkt.
EN: The raincoat's hanging up.

188
BE: Si het ehrei Haar pflägt.
ZH: Si hät ihri Haar pflägt.
EN: She brushed her hair.

189
BE: Di zwei Cheng hei glacht.
ZH: Di zwei Chind händ glacht.
EN: The two children are laughing.

190
BE: Dr Maa hät dr Schau abzogä.
ZH: Dä Maa hät de Schal abzogä.
EN: The man tied his scarf.

191
BE: D Bluomä se i dr Vasä.
ZH: D Bluämä sind i dä Waasä.
EN: The flower stands in a pot.

Appendix B

1
ZH: Ich han Freiziit.
BE: I ha Freiziit.
EN: I have spare time.

2
ZH: Ich han käi äigeni Band.
BE: I ha kei egeti Band.
EN: I do not have a band myself.

3
ZH: Mini Mueter isch ä no nie z Wien gsi.
BE: Mini Mueter isch o no niä z Wien gsi.
EN: My mother has never been to Vienna either.

4
ZH: Si liit det am Bode.
BE: Si ligt dert am Bode.
EN: She is lying there on the floor.

5
ZH: Ich wäis nöd werum si so abglänkt isch.
BE: I weis nid werum si so abglänkt isch.
EN: I don't know why she is so distracted.

6
ZH: Mit em Zug sälber zwänzg Minute.
BE: Mit em Zug säuber zwänzg Minute.
EN: With the train itself, twenty minutes.

7

ZH: Das git's äigentlich i käinere andere Spraach.

BE: Das git's eigentlech i kenere angere Spraach.

EN: This doesn't actually exist in any other language.

8

ZH: Ich mach vor alem vegetarisches Sushi.

BE: I mache vor auem vegetarisches Sushi.

EN: I mainly make vegetarian sushi.

9

ZH: Mäischstens ladet mich min Papi ii.

BE: Meischstens ladet mi mi Papi ii.

EN: Usually my dad invites me.

10

ZH: Bi de Spraache han i mir äifach nöd chöne vorschtele,
was i nachher wett mache.BE: Bi de Spraache han i mir eifach nid chönne
vorschteue, was i nachhär wott mache.EN: I just couldn't imagine what I wanted to do with
languages in the future.

11

ZH: Es gaht ja immerhin e chli um en grössere Betrag.

BE: Es geit ja immerhin e chli um ne grössere Betrag.

EN: We're dealing with a larger amount of money after all.

12

ZH: Ich find's schön, dass d Familie so zäme chunt.

BE: I finge's schön, dass d Familie so zäme chunt.

EN: I think it's nice that the family meets up like that.

13

ZH: Ziitig isch wahrschijnlijk informatiiver.

BE: Ziitig isch wahrschinglech informatiiver.

EN: Newspapers are probably more informative.

14

ZH: Aber s Problem isch, dass er grad kippt.

BE: Aber ds Problem isch, dass är grad kippt.

EN: But the problem is that he is about to tip over.

15

ZH: Politik hät mich au früener scho sehr intressiert.

BE: Politik het mi o früeher scho sehr intressiert.

EN: Politics has already interested me back in the days.

16

ZH: I de letschte zwäi Jahr bin ich vil go räise.

BE: I de letschte zwöi Jahr bin i viu ga reise.

EN: I did a lot of traveling over the past two years.

17

ZH: Si händ sich dänn entschide.

BE: Si hei sich denn entschide.

EN: They then took a decision.

18

ZH: D Frau staht am ne Fänschter.

BE: D Frou steit am ne Fänschter.

EN: The woman stands in front of a window

19

ZH: De Bueb staht uf em Schtuel und will ebe di
Cookies schtähle.BE: Dr Bueb steiht uf em Schtueu u wott äbe die
Cookies schtähle.

EN: The boy stands on the chair and wants to steal these cookies.

20

ZH: Am Wuchenänd bin i mal churz wägg gsi.

BE: Am Wuchenänd bin i mau churz wägg gsi.

EN: I left for a short time, on a weekend.

21

ZH: Im Huus, won i gwohnt han, hät's so es

Indonesischs Restaurant ghaa.

BE: Im Huus, won i gwohnt ha, het's so es Indonesischs

Restaurant ghaa.

EN: There was this Indonesian restaurant in the house
where I lived.

22

ZH: Ich han äigentlich wele i de Schwiiz bliibe.

BE: I ha eigentlech wöue ir Schwiiz bliibe.

EN: I actually wanted to stay in Switzerland.

23

ZH: Es isch äigentlich di ganz Wuche so gange.

BE: Es isch eigentlech di ganz Wuche so gange.

EN: This actually continued during the whole week.

24

ZH: De ganz Überfall wird äigentlich dadurch gschtöört,
dass öpper versuecht bi de Iigangstüür ine z choo.BE: Dr ganz Überfau wird eigentlech derdür gschtöört,
dass öpper versuecht bi dr Iigangstüür ine z choo.EN: The entire robbery is disturbed by someone trying
to enter the front door.

25

ZH: Jee nach dem versuecht me halt gwüssi

Chartekombinatione z haa.

BE: Jee nach däm versuecht me haut gwüssi

Chartekombinatione z haa.

EN: Depending on the situation, one tries to have
specific card combinations.

26

ZH: Es git sicher vil Aawändige.

BE: Es git sicher viu Aawändige.

EN: There are certainly many applications.

27

ZH: Es chönt irgend es Signal sii.

BE: Es chönt irgend es Signau sii.

EN: It could be any type of signal.

28

ZH: S linguistische Wüsse, wo nöötig isch, haltet sich
äinigmassen im Rahme.BE: Ds linguistische Wüsse, wo nöötig isch, hautet sech
einigmassen im Rahme.

EN: The linguistic knowledge, that is needed, is quite limited.

29

ZH: Grundsätzlich wär's scho cool, wänn i öppis chönt
mache mit Spraache.BE: Grundsätzlech wär's scho cool, wenn i öppis chönt
mache mit Spraache.

EN: In principle it would be cool if I could do something with languages.

30

ZH: Normalerwiis han i vier Voorläsige.

BE: Normalerwiis han i vier Voorläsige.

EN: Usually I have four lectures.

31

ZH: Japan bruucht ja vor alem im Summer vil Elektrizität.

BE: Japan bruucht ja vor auem im Summer viu Elektrizität.

EN: Japan needs a lot of electricity, especially in the summer.

32

ZH: Das isch wüerkli super toll gsi.

BE: Das isch wüerklech super toll gsi.

EN: That was really very nice.

33

ZH: De Wääg drüü isch nöd unbedingt für mich zuegschnitte gsii.

BE: Dr Wääg drüü isch nid unbedingt für mi zuegschnitte gsii.

EN: Path number three wasn't really tailored for me.

34

ZH: Wänn öpper mich grüeft hät, de bin i natürli grad uufgwacht.

BE: Wenn öpper mi grüeft het, denn bin i natürlich grad uufgwacht.

EN: If someone called me I naturally woke up right away.

35

ZH: Irgendwie hät no äine wele i d Bank ine.

BE: Irgendwie het no eine wöue i d Bank ine.

EN: Somehow, someone also wanted to enter the bank.

36

ZH: De Bueb git em Mäitli di Guezli abe.

BE: Dr Bueb git em Meitli d Guezli abe.

EN: The boy gives the cookies down to the girl.

37

ZH: Ich bi äigentlich däa wo dänn d Reschte isst.

BE: I bi eigentlech däa wo de d Räschte isst.

EN: I'm actually the one who eats the leftovers.

38

ZH: Ich ha scho jeenschti Aafraage gha.

BE: I ha scho jeenschti Aafraage gha.

EN: I have already had all kinds of requests.

39

ZH: Ich bi irgendwie scho i allne mögliche Rüüm ine gsii.

BE: I bi irgendwie scho i aune mögliche Rüüm drinne gsii.

EN: I have already been in all kinds of rooms.

40

ZH: Ich bi wüerkli a allem interessiert.

BE: I bi wüerklech a auem intrressiert.

EN: I am really interested in everything.

41

ZH: Die Ungwüsshäit, was jetz usehnt bi dem Bild, das find ich äifach faszinierend.

BE: Die Ungwüsshheit, was jetz usehnt bi däm Biud, das fing i eifach faszinierend.

EN: This uncertainty, what will turn out from this picture, I just find this fascinating.

42

ZH: Mer luegt use in Garte.

BE: Me luegt use i Garte.

EN: One looks outside, towards the garden.

43

ZH: Ich mach's nie jede Tag gliich.

BE: I mache's niä jede Tag gliich.

EN: I never do it the same way every day.

44

ZH: Ich ha im Näbefach Informatik.

BE: I ha im Näbefach Informatik.

EN: I study software engineering as a minor.

45

ZH: Ich studiere im Hauptfach latinischi Spraach- und Literaturwüesseschaft.

BE: I studiere im Houptfach latinischi Spraach- u Literaturwüesseschaft.

EN: I study Latin linguistics and literature as a major.

46

ZH: Arabisch möcht i scho lang mal belege.

BE: Arabisch möcht i scho lang mau belege.

EN: I have been wanting to take Arabic for a long time.

47

ZH: Ich mues säge, ich ha nöd so en Bezug zu dene Spraache.

BE: I mues säge, i ha nid so ne Bezug zu dene Spraache.

EN: I have to say, I do not have a strong attraction towards these languages.

48

ZH: Früener oder spööter wird ich mich wahrscheinlich scho mal draa waage.

BE: Früeher oder spääter wärd i mi wahrscheinlich scho mau draa waage.

EN: Sooner or later I will probably dare to do it.

49

ZH: I bi chuum jee überhaupt im Usland gsi.

BE: I bi chuum jee überhoupt im Usland gsi.

EN: I have barely been to foreign countries.

50

ZH: Maskierte Persone händ sich Zuetritt i de Bank verschafft.

BE: Maskierte Persone hei sech Zuetritt i d Bank verschafft.

EN: Masked people gained access to this bank.

51

ZH: Ich ha jetz äifach gmerkt, dass's mega ufwändig isch.

BE: I ha jetz eifach gmerkt, dass's mega ufwändig isch.

EN: I have simply noticed that it is very costly.

52

ZH: Es erschännt grad i welere Ziitschrift oder welere
Ziitig, dass es gsii isch.

BE: Es erchennt grad i welere Ziitschrift oder welere
Ziitig, dass es gsii isch.

EN: It recognizes right away in which magazine or paper
it appeared.

53

ZH: Sit dem ich das sälber mache, so vertäile, nimm
ich's ä vo dene Lüüt wänn si's mir vertäiled.

BE: Sit däm i das säuber mache, so verteile, nimm i's o
vo däne Lüüt wenn si's mir verteile.

EN: Ever since I do this myself, such distributions, I also
take them from people who distribute them to me.

54

ZH: Du wirsch scho nach Stundelohn zahlst.

BE: Du wirsch scho nach Stundelohn zaut.

EN: You are paid in hourly wages.

55

ZH: Ich find äifach New York e super Stadt.

BE: I finge eifach New York e super Stadt.

EN: I just think New York is a great city.

56

ZH: Uf jede Fall sind alli Bankaagschtelten am Bode
gläge.

BE: Uf jede Fau si aui Bankaagschteute am Bode gläge.

EN: In any case, all bank employees were lying on the floor.

57

ZH: S git so es paar Sache won i nach Rezäpt mache.

BE: S git so es paar Sache won i nach Rezäpt mache.

EN: There are a few things for which I use recipes.

58

ZH: Si tuet irgend en Täller poliere.

BE: Si tuet irgend es Täuer poliere.

EN: She is polishing some plate.

59

ZH: Wänn ich us mim Huus us chum, mues ich links
relativ e langi Straass abe.

BE: Wenn i us mim Huus use chume, mues i links relativ
e längi Straass abe.

EN: When I leave my house I have to go down a pretty
long street to the left.

60

ZH: S isch äifach en Schpraachkurs.

BE: S isch eifach e Schpraachkurs.

EN: It's simply a language course.

61

ZH: Ich ha rächt lang Tennis gschpilt.

BE: I ha rächt lang Tennis gschpiut.

EN: I played tennis for quite a long time.

62

ZH: Ich ha gäg äini gschpilt wo jetz Wältnummer acht
isch.

BE: I ha gäg eini gschpiut wo jetz Wäutnummer acht
isch.

EN: I played against someone who is now world number
eight.

63

ZH: De Trainer säit dir was du uf em Platz musch
mache.

BE: De Trainer seit dr was du uf em Platz muesch
mache.

EN: The coach tells you what to do on the court.

64

ZH: D Mueter isch näbetraa am abwäsche.

BE: D Mueter isch näbetraa am abwäsche.

EN: The mother is doing the dishes by the side.

65

ZH: Was mir wüerkli voorschwebt isch en akadeemisch
Karriere.

BE: Was mir wüerklech vorschwäbt isch e akadeemisch
Karriere.

EN: What I really have in mind is an academic career.

66

ZH: Di andere nüün mached d Vertüüfig imene andere
Beräich.

BE: Di angere nüün mache d Vertüüfig imene angere
Berich.

EN: The other nine focus on another subject.

67

ZH: Mir sind mal na es Semeschter in Münche gsii.

BE: Mir si mau no es Semeschter ds Münche gsii.

EN: We also went to Munich for a semester.

68

ZH: Ich han scho vorher im Film gschaft im e andere
Beräich.

BE: I ha scho vorhär im Fium gschaft im ne angere
Berich.

EN: I have worked with films before in a different area.

69

ZH: All die Lüüt won ich äigentlich scho kännt ha, die
Regisseure, die händ sozäge scho ihri Kamera-Lüüt ghaa.

BE: Au die Lüüt won i eigentlech scho kennt ha, d
Regisseure, die hei sozäge scho ihri Kamera-Lüüt
ghaa.

EN: All these people whom I already knew, these stage
directors, already had their cameramen.

70

ZH: Ich wäiss gar nümme.

BE: I weiss gar nümme.

EN: I don't remember.

71

ZH: Ich hoff er schlaat de Chopf nöd aa a de Kante.

BE: Ig hoffe är schlaat dr Chopf nid aa a de Kante.

EN: I hope he doesn't hit his head on the edge.

72

ZH: Du chasch mit jedem Spieler nur e gwüssi Aazahl
Fäldli laufe pro Rundi.

BE: Du chasch mit jedem Schpieler nume e gwüssi
Aazau Fäudli loufe pro Rundi.

EN: With every playing piece you can only go a certain amount of fields per round.

73

ZH: Ich han dihäi es chliises Schtudio.

BE: I ha deheime es chliises Schtudio.

EN: I have a little studio at home.

74

ZH: Di gröschte Fortschritt machsch äigentlich sälber.

BE: Di gröschte Fortschritt machsch eigentlech säuber.

EN: The largest progress, you actually make yourself.

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